



Good enough? A comparison of different harmonization procedures and their substantive consequences using the example of life satisfaction

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Abstract

Survey data harmonization can greatly improve the analytical potentials of survey data by making divergent measurements of the same construct comparable. While there may often be ideas on what is the optimal harmonization strategy (e.g. imputation or equipercentile equating), these approaches are not always feasible e.g. due to data-based restrictions or the lack of a suitable reference sample. Therefore, sometimes there is only a set of second-best strategies, such as linear equating or linear stretching, with no clear idea which of these should be given preference. The present paper takes this situation as a starting point and investigates the substantive consequences of different alternatives for a typical harmonization scenario using real-world survey data. Divergent substantive consequences are understood as differences in obtained analytical result patterns. Do the actual scientific conclusions differ when different harmonization approaches are employed? Results based on a variety of regression scenarios indicate that most substantive conclusions are obtained regardless of the harmonization approach chosen, but that harmonization procedures entailing an information loss due to recoding variables into smaller sets of categories should be avoided. Overall, this suggests that research based on sub-optimally harmonized scales can also yield valid scientific insights.

Keywords Harmonization · Stretching · Equating · Life satisfaction · Research data management

1 Introduction

While the comparability of divergent measurements across surveys has received attention for several decades (Dubrow and Tomescu-Dubrow 2016; Kolen and Brennan 2014; Kolen and Whitney 1982; Lord 1965; Singh et al. 2024), the social and behavioral sciences have only recently developed a vigorous interest in the issue of harmonizing dissimilar data sources (Tomescu-Dubrow et al. 2024; Wysmulek et al. 2022), with data harmonization

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understood as making comparable divergent measurements of the same concept. This encompasses dealing with divergences in cross-national data where data from different countries need to be harmonized, but it can also refer to longitudinal data in which e.g. the scale of a variable was revised between survey waves, creating a structural break and thus also a harmonization problem (Dubrow and Tomescu-Dubrow 2016; Granda et al. 2010; e.g. May et al. 2021; Pettinicchio and Maroto 2021; Tomescu-Dubrow and Slomczynski 2016). In line with this emerging interest, there are now many research projects that seek to harmonize divergent data that were not originally designed to be analyzed jointly using a variety of approaches in order to enhance the usefulness of these data (Biolcati et al. 2022; e.g. May et al. 2021; Schulz et al. 2021; Tomescu-Dubrow and Slomczynski 2016). This can also be seen as an outgrowth of the general increase survey data has seen in the last two decades: As more and more datasets are released, analyzing them jointly and thus enhancing the geographic and/or temporal scope of the analyses becomes more and more feasible and fruitful. However, methodological investigations into harmonization are still few and far between (Slomczynski et al. 2024), contrasting the booming field of survey data harmonization.

The present study takes the perspective of ex-post harmonization of divergent data sources as its starting point in order to compare the substantive consequences of a variety of approaches that could be taken here. The emphasis here is on the aspect of single-scale harmonization, i.e. of taking one questionnaire item and transforming it into a measure of the same meaning but different scale as found in another questionnaire (Kolen and Brennan 2014; Singh 2022). Moreover, this study is focused on harmonization within a single country (Germany), as the cross-national context exhibits a variety of additional complications that overlay the actual harmonization procedures (Slomczynski et al. 2024). For the purposes of this study, a measure of life satisfaction constitutes the core concept that will be harmonized. However, the research objectives are of a purely methodological nature and the study does not seek to contribute substantially to the life satisfaction literature.

A primary objective of data harmonization is making comparable the various components of several divergent datasets, and for this there is a variety of approaches of varying sophistication. Some of these are not always feasible, which necessitates recourse to second-best approaches. Moreover, some of these are based on a reduction of categories, thus inducing an information loss. This choice problem raises the question of the consequences of using such harmonizations. Therefore, comparing several such approaches is the core objective of the present study. For these approaches, the substantive consequences in the form of broad result patterns are investigated and compared to a benchmark approach. To what extent would divergent harmonization procedures actually lead to different scientific conclusions, i.e. when looking at the conclusion section of a study? Can less-than-optimal harmonization approaches still be viewed as “good enough”?

More precisely, the empirical approach taken to answer these research questions is to compare regression result patterns obtained for a total of seven different harmonization variants and a variety of additional model scenarios. These include models of varying size in terms of the number of variables included, using the harmonized variable as a dependent variable and as an independent variable, and varying whether a harmonization control is entered into the models. This yields a total of 96 models for which result patterns will be compared, some of which also include squared terms and interactions in order to also investigate the stability of such multiplicative terms in regression analyses.

In the end, the finding patterns obtained for these scenario combinations could imply that divergent strategies of harmonization lead to roughly similar results. In this case, researchers can more confidently apply data that would, from a purely methodological

standpoint, be regarded as sub-optimal. This would also have relevance for judging publications that are based on such harmonization procedures. In such a substantive result scenario, the implication for data quality would be that even a less-than-perfect harmonization can lead to valid results. From a data producer's point of view, this might also be used in planning more cost-effective survey harmonization efforts under circumstances of limited resources (Singh et al. 2024). Conversely, should the results diverge notably, this would imply that the data quality would benefit from focusing on more sophisticated methods. Researchers would then be well-advised to add different harmonization strategies to their robustness checks, and for data producers this would suggest prioritizing more harmonization resources at the relevant planning stages.

The results presented below broadly support the first view that even sub-optimally harmonized variables can lead to valid substantive conclusions: The correlations between the different versions are relatively large, and the divergences in result patterns between the regression models often are not of substantive relevance regarding the scientific conclusions that can be drawn. This is especially the case for those models that include an additional harmonization control, since such controls address certain shortcomings of the less-than-optimal harmonization approaches. In sum, this implies that research based on only imperfectly harmonized data nonetheless can usually be trusted. However, this is much less so for approaches that are based on arbitrarily recoding the variables into versions with fewer categories than present in the source variables: It appears that the information loss incurred when taking this approach leads to instability in the estimates, especially so for squared terms and interactions. Simplifying the source variables therefore appears to be an approach that should be avoided unless there are additional substantive considerations that mandate taking this route.

2 Background and state of the art

The broad methodological issue of data harmonization has seen increasing interest in recent years. Starting from an early focus on harmonizing test scores in order to compare test subjects across dissimilar survey designs (Kolen and Whitney 1982; Lord 1965), present-day harmonization efforts are often more strongly interested in adding new data to a pre-existing dataset or combining divergent data sources (May et al. 2021; Schulz et al. 2021; Tomescu-Dubrow and Slomczynski 2016). This can imply cumulating repeated cross-sections or dealing with individual-level panel data in which instruments have been revised over time. It can also imply adding datasets from other countries in order to expand the geographical scope of a dataset. What is common to all of these scenarios is that some form of harmonization often needs to be conducted in order to make divergent measures of the same theoretical concept comparable. Often this involves a distinction between target and source variable formats, where the latter is transformed to conform to the former. This distinction also is the foundation of this study, as I focus on the simple case of harmonizing one variable in two datasets.

Doing so I will focus on different types of harmonization procedures. The first and most sophisticated approach is equipercentile equating (Kolen and Brennan 2014; Singh 2022). The basic notion here is to transform the original scores of the variable to be harmonized in such a way that the distributional properties, i.e., aspects of the cumulative response distribution, of the target variable are mirrored (ibid.). This implies a potentially non-linear transformation that addresses potentially different response behaviors. More precisely, for

each percentile rank of the source variable, the value at the corresponding percentile rank of the target variable is assigned. In the case of discrete measures such as the ones investigated here, a linear interpolation is performed for the percentile ranks between the measured values.

The second approach that will be investigated here is linear equating. This approach transforms the source variable into a version that has the target variable's mean and standard deviation. As such it is suited only for normally distributed variables. Given that our focal variable of life satisfaction has a skewed nature, this is not the best option for harmonization, and it will be interesting to see how this approach then fares in comparison to the other approaches.

Third, a linear stretching of a variable involves harmonizing the minimum and maximum values of the source variable to that of the target variable, at times converting them to percentages (Cohen et al. 1999). For example, transforming a five-point scale into ten points results in the values of 1, 3.25, 5.5, 7.75, and 10. The drawback here is that any distributional differences between the variables, such as differential skewness, will not be addressed by this transformation. Moreover, it can be argued that the substantive meaning of the minimum value '1' differs for respondents when the maximum value is '5' instead of '10' (Tomescu-Dubrow and Slomczynski 2016; Weijters et al. 2010). Therefore, an adjustment of the linear stretch has been proposed that moves the source points into scale intervals, thus forgoing the exact mapping of the minimum and maximum (Cichocki and Jabkowski 2022; Tomescu-Dubrow and Slomczynski 2016). In the present example, this would mean that there are the values 1.5, 3.5, 5.5, 7.5 and 9.5 in the modified version of the five-to-ten-point stretch. The rationale is that when there are only five response options, these values capture a broader range of any latent variable than is the case when there are ten response options. This is reflected in the slightly divergent values for the minimum and maximum which mirrors the expansion of these ranges.

A final set of approaches that will be investigated in this study is related to the idea of reducing categories into e.g. "low" "medium" and "high" life satisfaction in order to make scales comparable. Such a recoding usually is necessarily arbitrary, as the cutoffs for this can be chosen in a number of ways. Furthermore, as noted above, the variable to be harmonized has a somewhat skewed nature, and dealing with this skewness when choosing the recategorization mapping also can be done in a number of ways. For this reason, three types of such recategorizations will be included, one of which will ignore the skewness of the variables, and two that will approach it in a different manner. The exact manner in which this is done is detailed graphically in the section "[Data and methodological approach](#)" alongside a corresponding depiction of the other harmonization approaches as well.

The analysis will thus include seven different harmonizations of life satisfaction and look at their substantive consequences in the form of regression model result patterns. Analyses like this are still relatively rare, i.e., there are only a few studies that pursued a comparison of harmonization alternatives with a similar real-world-data approach. A notable example is Cichocki and Jabkowski (2022) who investigated the consequences of stretching extremely dissimilar scales, i.e. stretching two points to eleven points. Their general conclusion is that this is not advisable. However, as noted above, they also pursued a modification of the linear stretch that moves the original scale points into intervals between the original scale points (see also Tomescu-Dubrow and Slomczynski 2016). That is, the two-point scale (0,1) was rescaled to (0.25, 0.75), and the eleven-point scale ranged from 0.05 to 0.95. This can ameliorate bias at least to some extent, at the cost of having dissimilar variable ranges (Cichocki and Jabkowski 2022, S. 3741). However, Batz et al. have

shown that linear stretching can be potentially also problematic for less divergent scales (Batz et al. 2016). In contrast, equipercentile equating is preferable in most single-item scale situations, as it provides less biased means due to the fact that distributional divergences are mitigated (de Jonge et al. 2014; Singh 2022).

Finally, there is also the general argument that response formats matter for response behavior (Alwin & Krosnick 1991; Cabooter et al. 2016; Tourangeau et al. 2000; Weijters et al. 2010). Insofar divergent response behaviors e.g. due to divergent difficulties implies divergent distributions etc., linear transformation methods or recodes that neglect such differences need to be seen as inferior also for this reason. Approaches that adjust for such differences, such as equipercentile equating, should be regarded as superior (Kolen and Brennan 2014; Singh 2024; Singh and Roth 2022; also see Tong and Kolen 2005). Therefore, my analytical setup will use equipercentile equating as the benchmark case with which the six other types of harmonization will be compared.

The analyses will also include a study dummy as a further harmonization control (Batz et al. 2016; Slomczynski et al. 2022). Given the data structure, this adjusts the means between the two included survey samples. This dummy then adjusts not only for potential methodological differences, but also for mean-related divergences between the harmonizations. It therefore also removes mean bias from the harmonization approaches, thus mitigating an important shortcoming of some of the harmonization approaches (Singh 2024). Put differently, adding a dummy adds an additional type of model-based harmonization on top of the variable-based one. Therefore, models with and without this dummy will be presented in order to expose more directly the mean-related consequences of the harmonization procedures, but it is not possible to determine what fraction of the dummy's effect is due to which type of divergence between the datasets.

I will look at life satisfaction as an outcome variable, but also as an explanatory variable with self-rated health as the outcome variable. However, the approach taken here is only loosely based on prior research of life satisfaction and self-rated health. As such, it is not the aim of the analyses to actually contribute new substantive insights to these two fields. The models rather intend to capture the most typical explanatory variables and covariates identified from the literature in order to mimic an actual substantive research interest. In this sense, the analyses are more complex than a purely stylized or simulation-based approach would probably allow. As detailed below, this includes a host of standard controls and demographics, but also more specialized variables derived from these literatures, such as church attendance and generalized trust. Moreover, some of the models also include interaction terms and squared terms, therefore allowing for an investigation of such non-linear relationships and their stability. The models therefore are reasonably complex in order to allow to adjudicate between the different harmonization approaches in a way that emulates the complexity of actual research models. I will provide more detail on them in the next section.

3 Data and methodological approach

This study seeks to test the consequences of harmonization procedures for substantive results that are based on “real-world” datasets. Therefore, a suitable combination of data that can be harmonized without too many assumptions about the geographical and/or temporal spacing of the studies is necessary. The German samples of the European Values Study 2017 (henceforth EVS, EVS 2022) and the International Social Survey Programme

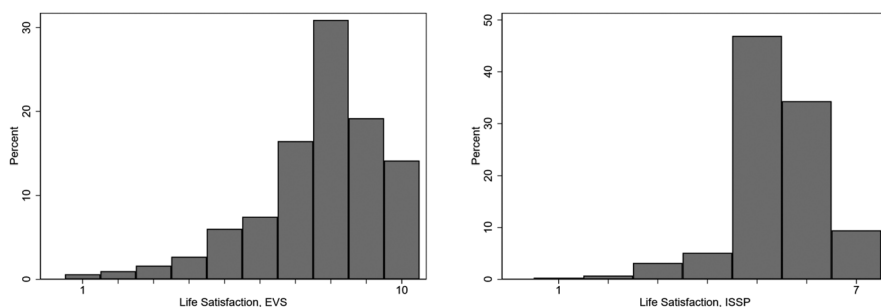


Fig. 1 Histograms for both life satisfaction variables

2018 (henceforth ISSP, ISSP Research Group 2019) satisfy this to a considerable extent.¹ They therefore provide the samples for this study.

The central variable of interest is life satisfaction. In the EVS it is measured as a ten-point-scale ranging from “dissatisfied” to “satisfied”, and the question text reads “All things considered, how satisfied are you with your life as a whole these days?”. In the ISSP it is measured with seven scale points with “completely dissatisfied” to “completely satisfied”, as endpoints, and the question text is “All things considered, how satisfied are you with your life as a whole nowadays?”, i.e. almost identical to the EVS text, thus comparability should be less of an issue than with more divergent wordings (Singh et al. 2023). The harmonization of these scales will be the primary empirical aspect of the study. As Fig. 1 shows for both source variables, life satisfaction is skewed in Germany, indicating relatively high levels of satisfaction. It is furthermore evident that the middle range of the ISSP is slightly less populated than is the case for the EVS, which suggests a difference in the way the scales are used by the respondents.²

As noted above, the analyses will include seven different forms of addressing this harmonization problem. These can be classified into two types of harmonization: One that seeks to transform the scale points in ways that preserve the original number of categories, and one that categorizes respondents, trading information loss for a harmonized scale range progression of the variables in the form of e.g. high, medium and low life satisfaction. More precisely, the following seven approaches are taken, as described in the chapter above:

1. Equipercntile equating: Aligns distributions
2. Linear equating: Aligns mean and standard deviation
3. Linear stretch: Aligns minimum and maximum
4. Modified linear stretch: Uses midpoints between values
5. Three categories, arbitrary
6. Four categories, arbitrary
7. Three categories, equal sizes

¹ <https://europeanvaluesstudy.eu/>; <https://issp.org/>

² Preliminary residual analyses showed that there are 30 problematic outliers in terms of excessively large negative residuals in the group of respondents with the lowest level of life satisfaction in the ISSP sample needed to be removed. This reduced the effective range of the ISSP variable.

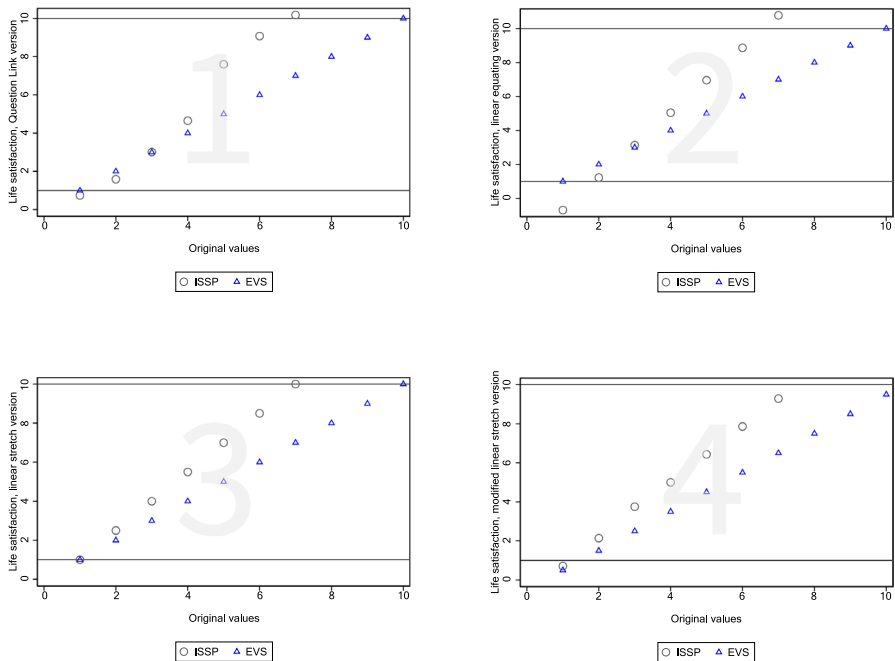


Fig. 2 Scatterplots depicting the four scaling/stretching harmonization procedures

Figures 2 and 3 depict the different approaches in the form of scatterplots and tables. In Fig. 2 the y-axes show the harmonized values, while the x-axes show the source values. The circles refer to the ISSP version and the triangles to the EVS version.

The first approach (1) refers to equipercentile equating (Kolen and Brennan 2014; Singh 2022). As described before, this approach is based on the use of reference samples (cf. also de Jonge et al. 2014) and takes into account distributional differences by allowing for a nonlinear transformation. The values taken here are derived from the QuestionLink engine (Singh and Roth 2022), which provides pre-calculated values for this approach³ and the datasets used. The non-linearity of this harmonization approach is visible in the scatterplot when looking at the middle range, which conforms to the divergences in the histograms shown in Fig. 1. In contrast, the linear equating approach (2) simply aligns the means and standard deviations. While this approach is feasible for normally-distributed variables and not for skewed variables such as the ones included here, I nonetheless will investigate its performance vis-à-vis the other approaches. As can be seen, the scale endpoints diverge strongly for these two skewed variables after a linear equating has been performed. Thirdly, the linear stretch approach (3) harmonizes the minima and maxima of the variables. However, as noted above, since there are arguments that the meaning of e.g. the value 1 is not independent of the maximum value of the scale, an alternative mode (4) has been devised which assigns values that are mid-points of intervals between the original values. This allows slight deviations between

³ <https://www.gesis.org/question-link>

Variant 5: Three categories, arbitrary										
Original values	1	2	3	4	5	6	7	8	9	10
EVS	1			2				3		
ISSP	1		2			3				

Variant 6: Four categories, arbitrary										
Original values	1	2	3	4	5	6	7	8	9	10
EVS	1				2		3		4	
ISSP	1				2	3	4			

Variant 7: Three categories, equal size										
Original values	1	2	3	4	5	6	7	8	9	10
EVS	1							2	3	
ISSP	1				2	3				

Fig. 3 The three recategorization procedures

the minima and maxima, while the overall shape of the relationship is the same. Therefore, the aforementioned drawback of having divergent scale endpoints applies here also, but to a lesser extent than is the case for the linear equating approach. This also is so because the scales employed in this study are more similar than, e.g., a harmonization from five or less points to eleven points.

Next, I will also look at recategorization approaches that simply group respondents into low, middle, and high life satisfaction groups in Fig. 3. These are by their nature arbitrary, and I will look at three variants. A first one (5) simply seeks to divide the variables into three equidistant categories along the scale, with the middle category being the largest capturing any extra categories after dividing the scales by three. This approach completely ignores the skewness of the variables.

In contrast, a second recategorization approach (6) acknowledges this skewness by creating four categories and allowing for a broader low-satisfaction category spanning four values, which are followed by three upper categories that are equidistant for each scale. Finally, a variant based on three categories (7) that are designed to be of roughly equal size in terms of respondent numbers also will be investigated. As can be seen, despite its rather curious form of having only one scale point as the middle category, this most strongly mimics the benchmark case: It not only acknowledges the skewed nature of the source variables, but also treats both original scales in a strongly divergent fashion.

This closeness to the benchmark case also is evident when looking at the bivariate correlations in Table 1. The coefficient in the bottom left corner of the table shows that variant 7 indeed is closely related to the benchmark case ($r=0.944$). In general, the correlations are relatively high across the board, but less so for the recategorized versions. Table 1 therefore already suggests that the different harmonization outcomes should not lead to vastly divergent results in a multivariable regression framework such as the one employed here.

The analyses will be carried out via linear regression for the first four types of harmonization and ordered logistic regression for the last three. Since the interest here lies

Table 1 Spearman correlations between the seven harmonization variants (analysis sample)

Harmonization variant (Life satisfaction)	1	2	3	4	5	6	7
Equipercntile equating	1.000						
Linear equating	0.9352	1.000					
Linear stretch	0.9615	0.9903	1.000				
Modified linear stretch	0.9328	0.9976	0.9903	1.000			
Three cat., arbitrary	0.8738	0.8738	0.8823	0.8738	1.000		
Four cat., arbitrary	0.7937	0.9526	0.9175	0.9526	0.8075	1.000	
Three cat., equal size	0.9443	0.8668	0.9144	0.8668	0.7829	0.7068	1.000

Table 2 Harmonized covariate variables

Covariate	Harmonization
Age	In years, no harmonization necessary
Gender	Male/Female, no harmonization necessary
Education	ISCED, recoded into six categories
Employment	Recoded into three categories
Income ^a	Recoded into deciles (based on EVS thresholds)
Children	Recoded into dummy
Married/partnered etc	Recoded into dummy
Health	Five categories, equipercntile-equated via QuestionLink (see above)
Church attendance:	Recoded into five categories (decreasing frequency)
Generalized trust	Recoded into two categories (EVS source variable has two categories)

^aMay contain misclassifications of +/- one decile due to divergent thresholds

in the broadest sense only on substantively divergent conclusions, I am foremost interested in the signs and significances of the regression coefficients in the models.⁴ These models include a variety of predictor variables which are also harmonized in order to make the data jointly analyzable. However, these harmonizations are treated as fixed, i.e. only life satisfaction varies between the models. Table 2 details the harmonized variables and how they have been treated in the data.

These variables largely are standard control variables and demographics, but with church attendance and generalized trust, I also add additional substantively relevant factors derived from the respective literatures (Assari 2013; Hamamura et al. 2017; King 1990;

⁴ This also means that the standard p-value significance filter of 0.05 is applied. While there is a vibrant discussion regarding null-hypothesis significance testing (Benjamin et al. 2018; Bernardi et al. 2017; McShane et al. 2019), the overall impression is that the conclusion sections in the social sciences by and large have relied and still rely on broad results patterns when summarizing and discussing a study's findings. As the findings generated here could also serve to appraise published results, this approach therefore appears feasible. Quantifying differences between the coefficient sizes is challenging for several reasons, including the divergent scales of the dependent variables, especially with regard to a comparison between findings from linear regression models with those from the ordered logit regressions, as the coefficient metrics are not directly exchangeable. Such a comparison will therefore be restricted to the linear regression models and the ordered logistic regression models separately in this study.

Lim and Putnam 2010). This allows to investigate a more realistic set of predictor variables that mimics the complex models in published research more strongly, since both life satisfaction and self-rated health are outcomes that are subject to a variety of potential influences (Jylhä, 2009; Ormel et al. 1999). I will also include additional models that involve an interaction between generalized trust and life satisfaction in the models explaining health. While this article does not pursue this as a substantive question, there are resource-based argument to be made for such a moderation process: Both life satisfaction and trust were shown to have buffering properties when it comes to health (Hamamura et al. 2017; Kim et al. 2021; Schneider et al. 2011; Siahpush et al. 2008), i.e. they act as important health-related resources. The question now is if one can substitute the other, which would become visible in a negative interaction term, or whether they enhance each others usefulness, which would lead to a positive interaction term. Finally, the models will be presented with a varying number of predictors, as also is often done in regression analyses in the empirical literature.

At the same time, I will not attempt to distinguish between focal and so-called control variables; After all, each of the factors could be the focal aspect of a hypothetical empirical paper. For some specific research questions, the models then might have to be revised in order to arrive at unbiased estimates (Bartram 2021), i.e. for specific focal aspects or research objectives there might be some potentially inappropriate controls that should be reconsidered (Cinelli et al. 2022; Kohler et al. 2023; Lundberg et al. 2021). Again, a substantive or even causal interpretation of the results is not the intention of this study; instead, the overall patterns of associations and the extent to which they diverge between the different harmonizations are in the focus. Appendix table A1 displays the descriptive statistics for all variables included in the models.

The main analyses are conducted in two large steps. In the first step, life satisfaction is the dependent variable, and in the second step, it is an independent variable alongside the other covariate variables mentioned in Table 2. For this second step, self-rated health is the dependent variable. Looking at both perspectives is informative because in both cases, a harmonization of variables can become consequential, but probably much more so if the harmonized variable is a dependent variable. Moreover, as mentioned above, in each step we present an additional set of models that includes a study dummy (i.e. an EVS-dummy) which also addresses certain mean-related deficiencies of the harmonization approaches.

4 Analyses

Since the results are based on many models and I am only interested in the signs and significances, I opted for a graphical representation of the models that uses shaded squares to depict significantly positive vs. negative coefficients. This makes it easier to compare the overall result patterns. Readers interested in the coefficient sizes find the full models in Tables 3, 4, 5, and 6 below. Figure 4 depicts the results for the dependent variable case. Three sizes of models are displayed, and the seven columns for each model size denote the seven harmonization approaches investigated here, in the same sequence as in Table 1. The first column is presented with thicker outlines as it represents the benchmark case, equipercentile equating based on the QuestionLink engine (see above), with which the other approaches are compared. Deviations from this result set are therefore seen as an indication of the deficiency of the other six harmonization approaches when it comes to the substantive result patterns.

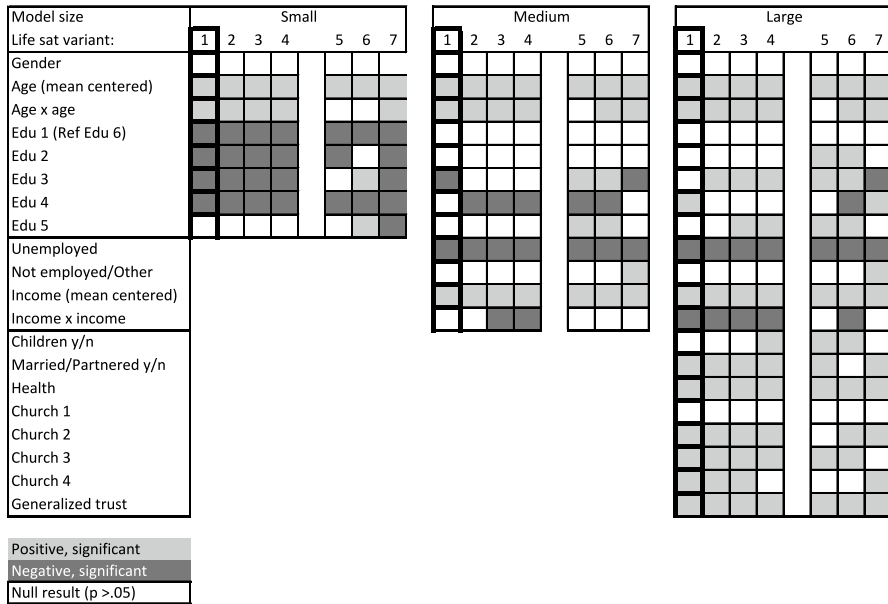


Fig. 4 Life satisfaction as dependent variable, without study dummy

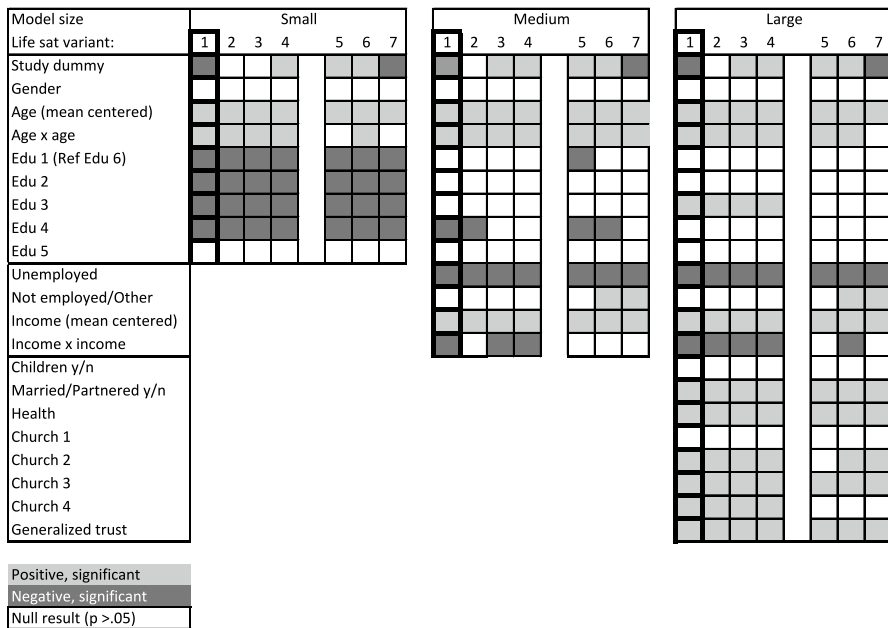


Fig. 5 Life satisfaction as dependent variable, with study dummy

Table 3 Exemplary models (life satisfaction as outcome, linear regressions, incl. study dummy)

	Life sat 1	Life sat 2	Life sat 3	Life sat 4
Gender	0.0172 (0.0506)	0.0136 (0.0523)	0.0126 (0.0480)	0.0126 (0.0472)
Age (mean centered)	0.0170** (0.00195)	0.0173** (0.00202)	0.0157** (0.00185)	0.0154** (0.00182)
Age × age	0.000349** (0.000105)	0.000357** (0.000109)	0.000327** (9.99e−05)	0.000320** (9.82e−05)
Edu 1 (Ref Edu 6)	0.144 (0.296)	0.126 (0.305)	0.0374 (0.280)	0.0157 (0.276)
Edu 2	0.169 (0.108)	0.157 (0.112)	0.148 (0.102)	0.146 (0.101)
Edu 3	0.207* (0.0856)	0.198* (0.0884)	0.185* (0.0811)	0.181* (0.0798)
Edu 4	0.104 (0.0794)	0.0931 (0.0819)	0.123 (0.0752)	0.128 (0.0740)
Edu 5	0.152 (0.0851)	0.153 (0.0878)	0.140 (0.0806)	0.137 (0.0793)
Unemployed	− 0.779** (0.169)	− 0.709** (0.174)	− 0.682** (0.160)	− 0.675** (0.157)
Not employed/other	0.0836 (0.0726)	0.109 (0.0749)	0.0964 (0.0688)	0.0946 (0.0677)
Income (mean centered)	0.0699** (0.0109)	0.0700** (0.0112)	0.0635** (0.0103)	0.0620** (0.0101)
Income × income	− 0.00974** (0.00347)	− 0.00939** (0.00359)	− 0.00931** (0.00329)	− 0.00921** (0.00324)
Children y/n	0.0783 (0.0606)	0.0775 (0.0626)	0.0813 (0.0574)	0.0824 (0.0565)
Married/partnered y/n	0.145* (0.0612)	0.162* (0.0632)	0.139* (0.0580)	0.134* (0.0571)
Health	0.665** (0.0293)	0.702** (0.0303)	0.622** (0.0278)	0.605** (0.0273)
Church 1 (Ref Church 5)	− 0.0333 (0.263)	0.102 (0.271)	0.0514 (0.249)	0.0418 (0.245)
Church 2	0.397** (0.115)	0.407** (0.118)	0.384** (0.109)	0.378** (0.107)
Church 3	0.204** (0.0634)	0.207** (0.0655)	0.191** (0.0601)	0.187** (0.0591)
Church 4	0.165** (0.0622)	0.149* (0.0642)	0.142* (0.0589)	0.139* (0.0580)
Generalized trust	0.348** (0.0535)	0.336** (0.0552)	0.316** (0.0507)	0.310** (0.0498)
Study dummy	− 0.270** (0.0625)	0.0312 (0.0645)	0.173** (0.0592)	0.271** (0.0582)
Constant	5.156** (0.143)	4.713** (0.147)	4.899** (0.135)	4.370** (0.133)
Observations	3,165	3,165	3,165	3,165
Adjusted R-squared	0.245	0.240	0.234	0.234

Table 3 (continued)Standard errors in parentheses. $**p < 0.01$, $*p < 0.05$

When looking at the first four models which are based on equating or stretching, there already is a relatively strong accordance in terms of the results patterns. While there are small deviations here and there, the overall conclusions based on these four models would be largely similar. This is the case for all three model size variants. The strong correlations between these harmonization approaches reported in Table 1 therefore are mirrored by this overall result pattern.

However, when looking at the last three models which are based on a reduction of the number of unique values of the variables, there is more variability. This variability also is evident for the squared term of income, which suggests that findings related to multiplicative predictor variables are less stable with regard to this specific type of harmonization approach. It furthermore appears that the instability of results based on re-categorizations is largely unrelated to model size, as these differences also appear to emerge in settings in which there are fewer predictors included in the models.

Figure 5 presents the same models, but with the above-mentioned addition of a dummy for the EVS sample. As expected, the results become more consistent across the different models than was the case in Fig. 1. This shows that the introduction of harmonization controls can, even in the minimal case of having just two datasets, overcome some of the limitations of harmonization procedures in that it produces the substantive result pattern that would be obtained using the most-preferred harmonization approach. The study dummy itself increases the adjusted r-squared by between 0.1 and 0.7 percentage points in the small models and 0.1–0.5 percentage points in the medium or large models.

Next, I turn to the findings for the independent variable case where self-reported health is the dependent variable. The logic of the result presentation is largely the same, but here there is an additional type of model which includes an interaction term between life satisfaction and generalized trust. The overall implications for this case are similar to the dependent variable case, albeit on a lower scale: In Fig. 6, the divergences are more pronounced in the models based on a reduction of answer categories, but on an overall lower level, since the differences here are only in one of the predictor variables. However, it is important to note that even in this case, the choice of harmonization approach matters. Regarding the interaction terms, it is significant only in the benchmark case, which suggests that results for this type of multiplicative model component also are sensitive to the type of harmonization employed.

For this perspective I also add a study dummy in order to investigate the consequences of adding this basic harmonization control, and the results are displayed in Fig. 7. There again is a reduction of variability in the result patterns, but of course from a lower level to begin with. It is especially evident for the approaches based on categorizations, where the study dummy also achieves statistical significance throughout. Therefore, based also on these models, it is evident that harmonization controls can establish more stable result patterns also when including the harmonized variable as a predictor variable. In the equating/stretch models the study dummy itself is seldomly statistically significant and explains only 0.1 percentage points of the variance (adjusted r-squareds), while it explains between 0.1 and 1.4 percentage points in the models based on categorizations.

In a final step I look briefly at the effect sizes in Tables 3, 4, 5 and 6 where the full models for both perspectives and the seven types of harmonization are displayed. These

Table 4 Exemplary models (life satisfaction as outcome, ordinal logit regressions, incl. study dummy)

	Life sat 5	Life sat 6	Life sat 7
Gender	– 0.0366 (0.082)	0.0269 (0.070)	0.00493 (0.072)
Age (mean centered)	0.0202** (0.003)	0.0201** (0.003)	0.0194** (0.003)
Age × age	0.000348* (0.000)	0.000396** (0.000)	0.000294 (0.000)
Edu 1 (Ref Edu 6)	– 0.401 (0.474)	– 0.0974 (0.426)	– 0.00975 (0.432)
Edu 2	0.226 (0.174)	0.136 (0.151)	0.281 (0.153)
Edu 3	0.224 (0.140)	0.145 (0.118)	0.163 (0.121)
Edu 4	0.0918 (0.126)	0.112 (0.109)	0.162 (0.112)
Edu 5	0.190 (0.139)	0.178 (0.117)	0.158 (0.121)
Unemployed	– 0.872** (0.280)	– 1.030** (0.252)	– 0.753** (0.267)
Not employed/other	0.229 (0.118)	0.210* (0.102)	0.286** (0.104)
Income (mean centered)	0.0884** (0.018)	0.0892** (0.015)	0.0836** (0.015)
Income × income	– 0.00506 (0.006)	– 0.00954* (0.005)	– 0.00185 (0.005)
Children y/n	0.0774 (0.099)	0.0697 (0.085)	0.0798 (0.086)
Married/partnered y/n	0.283** (0.098)	0.230** (0.085)	0.258** (0.086)
Health	0.942** (0.052)	0.917** (0.045)	0.861** (0.045)
Church 1 (Ref Church 5)	– 0.0812 (0.446)	0.292 (0.388)	0.0968 (0.389)
Church 2	0.345 (0.192)	0.467** (0.162)	0.383* (0.164)
Church 3	0.279** (0.103)	0.203* (0.088)	0.180* (0.089)
Church 4	0.120 (0.099)	0.138 (0.086)	0.106 (0.087)
Generalized trust	0.385** (0.086)	0.427** (0.074)	0.399** (0.075)
Study dummy	0.987** (0.102)	1.934** (0.093)	– 0.948** (0.090)
Cut 1	– 0.464 (0.268)	1.444** (0.207)	1.783** (0.204)
Cut 2	4.301** (0.250)	4.166** (0.214)	3.885** (0.215)

Table 4 (continued)

	Life sat 5	Life sat 6	Life sat 7
Cut 3		6.658**	
		(0.234)	
Observations	3,165	3,165	3,165
Pseudo R-squared	0.166	0.172	0.125

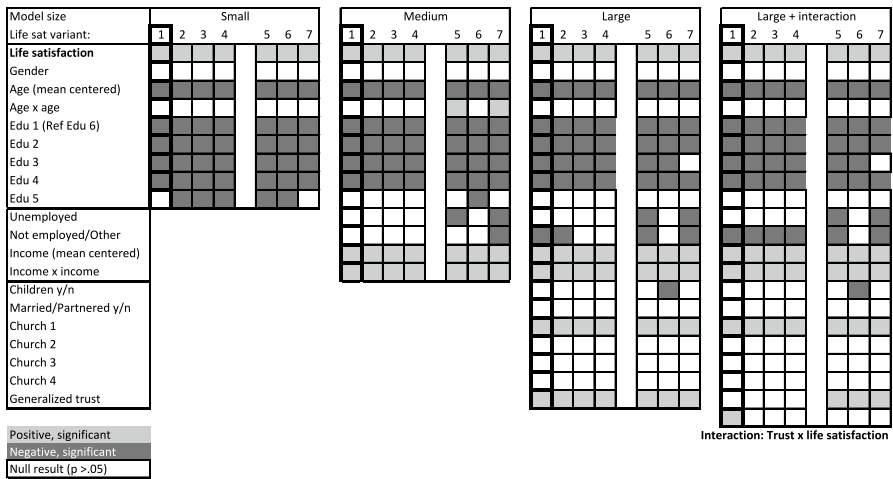
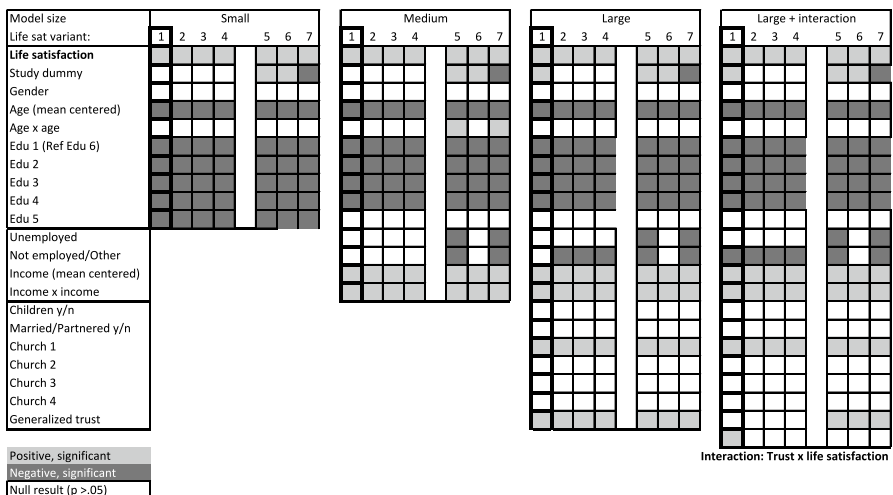
Standard errors in parentheses. ** $p < 0.01$, * $p < 0.05$ **Fig. 6** Life satisfaction as independent variable, without study dummy**Fig. 7** Life satisfaction as independent variable, with study dummy

Table 5 Exemplary models (health as outcome, linear regressions, incl. interactions and study dummy)

	Model 1	Model 2	Model 3	Model 4
Gender	– 0.0107 (0.0285)	– 0.0103 (0.0285)	– 0.0105 (0.0286)	– 0.0107 (0.0287)
Age (mean centered)	– 0.0169** (0.00107)	– 0.0169** (0.00107)	– 0.0169** (0.00108)	– 0.0169** (0.00108)
Age × age	6.92e–05 (5.95e–05)	6.72e–05 (5.93e–05)	7.04e–05 (5.96e–05)	7.16e–05 (5.97e05)
Edu 1 (Ref Edu 6)	– 0.439** (0.166)	– 0.433** (0.166)	– 0.419* (0.167)	– 0.415* (0.167)
Edu 2	– 0.230** (0.0608)	– 0.223** (0.0606)	– 0.226** (0.0609)	– 0.226** (0.0610)
Edu 3	– 0.166** (0.0482)	– 0.161** (0.0481)	– 0.162** (0.0483)	– 0.162** (0.0484)
Edu 4	– 0.196** (0.0446)	– 0.193** (0.0445)	– 0.203** (0.0447)	– 0.206** (0.0448)
Edu 5	– 0.0413 (0.0480)	– 0.0410 (0.0478)	– 0.0406 (0.0481)	– 0.0403 (0.0481)
Unemployed	– 0.136 (0.0954)	– 0.154 (0.0951)	– 0.153 (0.0956)	– 0.155 (0.0957)
Not employed/other	– 0.0824* (0.0409)	– 0.0853* (0.0408)	– 0.0847* (0.0411)	– 0.0845* (0.0411)
Income (mean centered)	0.0254** (0.00616)	0.0256** (0.00614)	0.0265** (0.00617)	0.0269** (0.00618)
Income × income	0.00610** (0.00196)	0.00610** (0.00196)	0.00625** (0.00197)	0.00627** (0.00197)
Children y/n	– 0.0524 (0.0342)	– 0.0521 (0.0341)	– 0.0541 (0.0343)	– 0.0546 (0.0343)
Married/partnered y/n	0.0267 (0.0346)	0.0244 (0.0345)	0.0276 (0.0346)	0.0287 (0.0347)
Church 1 (Ref Church 5)	0.370* (0.148)	0.346* (0.148)	0.360* (0.149)	0.364* (0.149)
Church 2	– 0.0528 (0.0648)	– 0.0540 (0.0646)	– 0.0543 (0.0649)	– 0.0535 (0.0651)
Church 3	0.00385 (0.0358)	0.00359 (0.0357)	0.00495 (0.0359)	0.00571 (0.0359)
Church 4	0.0265 (0.0351)	0.0299 (0.0350)	0.0300 (0.0352)	0.0304 (0.0352)
Life sat 1	0.194** (0.0115)			
Generalized trust	– 0.0219 (0.0634)	0.0394 (0.0584)	0.0287 (0.0624)	0.0532 (0.0541)
Trust × Life sat 1 (centered on 5)	0.0496** (0.0187)			
Life sat 2		0.196** (0.0114)		

Table 5 (continued)

	Model 1	Model 2	Model 3	Model 4
Trust×Life sat 2 (centered on 5)		0.0315 (0.0177)		
Life sat 3			0.208** (0.0123)	
Trust×Life sat 3 (centered on 5)			0.0365 (0.0197)	
Life sat 4				0.210** (0.0125)
Trust×Life sat 4 (centered on 5)				0.0350 (0.0200)
Study dummy	0.0845* (0.0353)	0.0180 (0.0351)	− 0.0162 (0.0354)	− 0.0390 (0.0356)
Constant	2.885** (0.0653)	2.944** (0.0642)	2.941** (0.0654)	3.060** (0.0630)
Observations	3,165	3,165	3,165	3,165
Adjusted R-squared	0.294	0.298	0.291	0.289

Standard errors in parentheses. ** $p < 0.01$, * $p < 0.05$

refer to linear or ordinal logit regressions with all predictor variables for both dependent variables, including study dummy and in the case of health as an outcome, the interactions terms. For the equating and stretching variants, in both the dependent and independent variable case, there are largely similar effect sizes for each of the significant predictors across these models. The differences that are visible are largely relatable to the divergent ranges of the dependent variables displayed in appendix Table A. This indicates that, at least for the linear regression case, results that are similarly significant also are of reasonably similar coefficient magnitude, further substantiating the overall finding that these four types of harmonizations lead to largely similar substantive conclusions.

The models based on a recategorization of the variable displayed in Tables 4 and 6, the results are not directly comparable to the linear case due to the different methodology and the divergent scalings of the life satisfaction predictor variable. However, they appear to be of similar magnitude. But we do see for the ordinal logit models that the study dummy changes its sign, which suggests that the different categorization schemes introduce divergent mean differences in the sample that become visible via this study dummy's sign changes.

Table 6 Exemplary models (health as outcome, linear regressions, life satisfaction as categories, incl. interactions and study dummy)

	Model 5	Model 6	Model 7
Gender	– 0.00390 (0.0290)	– 0.0130 (0.0286)	– 0.00680 (0.0290)
Age (mean centered)	– 0.0162** (0.00109)	– 0.0165** (0.00107)	– 0.0165** (0.00109)
Age × age	0.000104 (6.04e–05)	7.78e–05 (5.95e–05)	0.000107 (6.03e–05)
Edu 1 (Ref Edu 6)	– 0.381* (0.169)	– 0.396* (0.167)	– 0.423* (0.169)
Edu 2	– 0.221** (0.0618)	– 0.217** (0.0608)	– 0.246** (0.0618)
Edu 3	– 0.151** (0.0490)	– 0.152** (0.0482)	– 0.152** (0.0490)
Edu 4	– 0.189** (0.0454)	– 0.180** (0.0448)	– 0.205** (0.0453)
Edu 5	– 0.0318 (0.0487)	– 0.0403 (0.0480)	– 0.0347 (0.0487)
Unemployed	– 0.213* (0.0969)	– 0.133 (0.0958)	– 0.228* (0.0969)
Not employed/other	– 0.0908* (0.0416)	– 0.0798 (0.0410)	– 0.101* (0.0416)
Income (mean centered)	0.0306** (0.00624)	0.0258** (0.00615)	0.0298** (0.00624)
Income × income	0.00526** (0.00200)	0.00601** (0.00196)	0.00476* (0.00199)
Children y/n	– 0.0484 (0.0347)	– 0.0459 (0.0342)	– 0.0483 (0.0347)
Married/partnered y/n	0.0293 (0.0351)	0.0265 (0.0345)	0.0261 (0.0351)
Church 1 (Ref Church 5)	0.412** (0.150)	0.376* (0.148)	0.398** (0.150)
Church 2	– 0.00709 (0.0658)	– 0.0394 (0.0648)	– 0.0183 (0.0657)
Church 3	0.0125 (0.0364)	0.00392 (0.0358)	0.0231 (0.0363)
Church 4	0.0461 (0.0356)	0.0313 (0.0351)	0.0444 (0.0356)
Life sat 5: Cat 1	– 0.548** (0.131)		
Life sat 5: Cat 3	0.564** (0.0430)		
Life sat 6: Cat 1		– 0.495** (0.0749)	
Life sat 6: Cat 3		0.494** (0.0512)	
Life sat 6: Cat 4		0.786** (0.0611)	

Table 6 (continued)

	Model 5	Model 6	Model 7
Life sat 7: Cat 1			– 0.411** (0.0512)
Life sat 7: Cat 3			0.375** (0.0502)
Generalized trust	0.134** (0.0448)	0.122* (0.0533)	0.131** (0.0463)
Trust×Life sat 5: Cat 1	0.00850 (0.580)		
Trust×Life sat 5: Cat 3	0.0390 (0.0582)		
Trust×Life sat 6: Cat 1		– 0.0533 (0.139)	
Trust×Life sat 6: Cat 3		0.0325 (0.0679)	
Trust×Life sat 6: Cat 4		– 0.0104 (0.0791)	
Trust×Life sat 7: Cat 1			– 0.0114 (0.0776)
Trust×Life sat 7: Cat 3			0.0423 (0.0655)
Study dummy	– 0.0927* (0.0365)	– 0.252** (0.0382)	0.166** (0.0370)
Constant	3.204** (0.0624)	3.266** (0.0656)	3.339** (0.0642)
Observations	3,165	3,165	3,165
Adjusted R-squared	0.271	0.295	0.272

5 Discussion and conclusion

Data harmonization is an important tool that can be used productively to overcome divergences between data, thus enhancing their usefulness beyond their original intentions. This can be relevant in a longitudinal resp. comparative context where there is always a tension between cross-temporal resp. geographical consistency and innovation, but also when it comes to the more and more important perspective of merging divergent data that had not been designed with such a merge in mind. This latter case is important, as the social and behavioral sciences seek to capitalize further on the data that are available for analyses, be it for meta-analyses, or for the purpose of enhancing and enriching any data base potentially used. Dealing with measurement differences therefore also becomes an increasingly relevant precondition of these types of data use.

The present study focuses on the issue of single-item scale harmonizations. For this there exists a variety of approaches that have been described above, alongside prior arguments and evidence that equipercenile equating is suited particularly well for such a transformation. Generally, best-suited approaches often are relatively demanding, in this case necessitating the existence of a reference sample, which may not always be available.

Researchers therefore sometimes need to rely on less sophisticated or second-best methods. Moreover, published studies may also have relied on less-than-optimal harmonizations, which may have implications for their results. The present paper started from these considerations and sought to find out to what extent sub-optimal harmonization procedures result in divergent “real-world” results when compared to a benchmark harmonization approach. The benchmark case of equipercentile equating was compared to linear equating, two forms of linear stretching, and three types of recategorizations, for a total of seven harmonization variants.

The results based on a total of 96 models show that the choice of harmonization matters. This is important in that it suggests that whenever possible, there should be a recourse to more sophisticated methods. Put differently, time and resources put into the best-possible harmonization approach are a good investment. However, when looking across the result patterns, it becomes apparent that even sub-optimal harmonization procedures very often will lead to roughly similar result patterns. This implies that research based e.g. on a linear stretch is not dubious per se, and in circumstances where e.g. a suitable reference sample is not present and equipercentile equating is thus not possible, choosing this method may still provide relevant and trustworthy results. This is also evident when looking at the coefficient sizes for these models, which were of roughly similar size and varied expectably along with the divergent ranges of each of the harmonized variable.

The extent to which any differences emerge also is related to the role of the harmonized variable at hand: The divergences are less pronounced when the harmonized variable is an independent variable rather than the dependent variable in the models. This also shows that regression methods are somewhat robust against the type of error introduced by adding a less-than-optimally harmonized predictor variable. However, even here, there were differences regarding some of the predictors, so that in this case also, the harmonization approach should be chosen with care. This is even more so whenever the central substantive interest is focused on relationships that are modeled as multiplicative terms, such as squared terms or interactions, which is often the case.

However, not every type of harmonization can be seen as sufficient: Those approaches that are based on a recategorization, i.e., that induced a significant information loss via a reduction of categories, diverge more strongly from the benchmark case. Here it is more important to be cautious when results are based on this type of approach. This appears especially so for multiplicative terms such as squared terms. The conclusion here thus is to steer clear from this type of harmonization, if possible. This does not rule out that a recategorization can be substantively important, such as when converting household income into a poverty dummy, or dichotomizing age variables for some theoretical or substantive reason. Similarly, when dealing with substantive ordinal categories such as those often found in education variables, a simplifying recategorization sometimes cannot be avoided unless a more complex approach is taken, at times at the cost of predictive power due to a reduction of meaningful variance (Granda et al. 2010; Schneider 2010, 2022). Notwithstanding these cases, for rating scales such as the ones applied here, this approach should not be taken as a harmonization technique, at least not without additional recourse to other harmonization procedures, e.g. in a robustness check.

While the present study has taken important steps towards adjudicating between different types of harmonization approaches, there are several issues that need to be left for future research. For one, the harmonization problem at the center of this study, harmonizing seven to ten scale points, was quite expedient in that the scales were relatively similar to begin with. In the case of four vs. eleven points or three vs. seven points, the result patterns may differ more strongly. It therefore remains to be seen whether more difficult harmonization scenarios

in the form of more divergent source variables also can be approached with reasonable confidence using less-than-optimal harmonizations. Moreover, the base methodology here could be expanded to also include an SEM framework. That would also allow for more complex relationships for which divergent harmonization approaches may be more relevant than is apparently the case for linear and logistic regression. Related to this is the issue of harmonizations of multiple-item-scales, which would enable further types of harmonization approaches to be compared, such as IRT-based harmonizations (Lord and Wingersky 1984; Nissen et al. 2015). This could also include further comparisons with other purely model-based resolutions for data harmonization problems (Gangl 2023).

Furthermore, at present it must remain open to what extent the seeming similarity between the first four harmonization approaches can be generalized to other countries. For example, in countries with a less skewed distribution of life satisfaction, linear equating may per se be much more feasible than was the case for Germany and may therefore be a more valid approach. Therefore, it always appears useful to perform a variety of harmonizations and to test them against each other. However, given the details of the equipercntile equating approach outlined above, including other countries would require an extraction of specific equations for those countries. Such equations are presently not included in the QuestionLink database used here and deriving them is beyond the scope and interests of the present article.

A cross-country investigation would entail also dealing with the issue of cross-national comparability, which is not easily done for single-item harmonizations and may require additional measurements and conditions (Jovanović and Brdar 2018; Saris et al. 2018; Singh and Quandt 2024; Vandenberg and Lance 2000). Finally, the divergent methodologies necessary for the different harmonization approaches make it difficult to derive a numerical measurement of the result deviations (or bias in the effect strengths) introduced when compared to the benchmark case. This is inherent in the divergent transformations of these variables that renders such comparisons difficult. Therefore, I could only refer to the coefficient sizes for the two general types of harmonization approaches separately, for which the findings were in line with the general conclusions. This issue might, however, be approached in a simulation setting, but pursuing this is also well beyond the scope of the present article.

In sum, data harmonization in the context of single-variable divergences can be handled in a wide range of ways and with different advantages and disadvantages. While there are clear ideas on what would be the optimal procedure, that procedure cannot always be conducted. However, when it comes to the result patterns generated across the different types of harmonization, it appears that less-than-optimal procedures can be “good enough” to provide meaningful and reliable results as long as they are not based on simplifying recategorizations that induce information loss.

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Declarations

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